

Kinaawa High School – Mugongo

END OF YEAR ASSESSMENT 2025

PRINCIPAL MATHEMATICS 1

Time: 3 hours

Instructions: Attempt all items in Section A and any four in Section B. Use clear conclusions and show all necessary steps.

SECTION A

Item 1. A student who missed the lesson on differentiation has been given an assignment he is unaware of and has asked you to help him. The question is as below.

Differentiate $\sqrt{x}$ with respect to x from first principles.

Task: Help to show this student that the answer is $\frac{1}{2\sqrt{x}}$. (6 marks)

Item 2. An engineer for a village bridge has to find the angles he needs to put the poles holding the bridge concrete during its construction. He has given the equation

$$\cos(x + 20^\circ) - \cos(x + 80^\circ) = 0.5 \quad \text{for } 0 \leq x \leq 360^\circ.$$

Task: Determine the angles required to place the poles. (5 marks)

Item 3. A student was asked to use **Binomial Expansion** knowledge for expanding $(1 + x)^{\frac{1}{2}}$ up to the first three terms of the expansion.

Task: As a pure mathematics student, show clearly how it is done. (6 marks)

Item 4. A school disciplinary committee is to be set up comprised of 8 students. The school has chosen 6 girls and 8 boys who are well disciplined from which they have to form the committee. The Head of Disciplinary Committee prefers to have either 3 girls or 5 girls on the committee formed.

Task: Help the Head of DC determine the possible number of ways of achieving such from those available. (5 marks)

Item 5. A student was asked to find the area under the curve of the function:

$$f(x) = x^2 + 3$$

from $x = 1$ to $x = 2$.

Task: Use definite integration to solve this.

Item 6. A student is given an arithmetic progression (AP) where the first term is $a = 5$ and the common difference is $d = 3$. Find the 10th term of the AP.

SECTION B

(Pick any four items)

Item 7.

(a) An environmental scientist is designing a simple calculator program to estimate the percentage increase in greenhouse gas concentration over time. The formula used in the program is:

$$\ln(1 - 2x)$$

(i) Use Maclaurin's theorem to expand $\ln(1 - 2x)$ in ascending powers of x as far as the term in x^3 .

(ii) Use small changes to find the approximate value of $\sqrt[3]{1005}$, correct to 2 decimal places.

Item 8.

The alternating current in a circuit is given by:

$$I(\theta) = 12 \cos \theta + 16 \sin \theta$$

Tasks:

1. Express $I(\theta)$ in the form $R \cos(\theta - \alpha)$ by finding R and α .
2. Find the maximum current and the angle θ at which it occurs.
3. If safe operation requires $I(\theta) = 15$, find all values of θ in $0^\circ \leq \theta \leq 180^\circ$ that meet safety.

Item 9.

A cocoa cooperative's revenue function is given by

$$R(t) = \frac{3t^2 + 11t + 6}{(t+1)(t+2)(t+3)} \text{ million UGX,}$$

where t is the number of months after harvest.

Task:

1. Decompose $R(t)$ into partial fractions to analyze the monthly revenue components.

Item 10.

During a market research, the model

$$\sqrt{x+6} + \sqrt{4-x} = \sqrt{1-3x}$$

has been developed for the first new product X put forward by the firm, where x represents overall consumer reaction towards the new product.

The company analyst has been tasked to check, and if the verified output is negative, then the product should not be sold to the public.

Unfortunately, the analyst is too sick and asked you, the assistant, to help out but forgot to give the password to his computer. He left a clue word “ARRANGE” to access the three different user’s accounts he has.

Task:

1. Help determine whether the new product should be rolled out for sale or not by the company. (Hint: verify your answer.) (5 marks)
2. In how many possible arrangements in a row are there for the password to his accounts given that:
 - (a) User account 1: the two “R’s” are together.
 - (b) User account 2: the two “R’s” are not together.

(7 marks)

Item 11.

A company that manufactures plastic water tanks records its monthly profit $P(x)$ (in thousand shillings) as a function of the number of tanks produced x . The relationship is found to follow a cubic polynomial model:

$$P(x) = x^3 - 13x + k$$

Tasks:

1. Given that the polynomial $P(x)$ is exactly divisible by $(x - 4)$, find the value of k . Hence, solve the equation:

$$x^3 - 13x + k = 0$$

2. Solve the inequality:

$$\frac{x^2 - x - 18}{x + 3} \geq \frac{x}{2}$$

Item 12.

A telecom company is installing a new mast in a remote area near Fort Portal. They plan to run two parallel fiber optic cables to connect the mast.

The first cable, L_1 , has the equation:

$$3x - 4y = 12.$$

The second cable, L_2 , needs to pass through a relay station at $R(1, -2)$.

- (a) Find the gradient of the first cable, L_1 .
- (b) Find the equation of the second cable, L_2 .
- (c) What is the shortest distance from a village at $V(10, 8)$ to the first cable, L_1 ?

Item 13.

(a) A small irrigation project near Mukono requires calculating the area of a trapezoidal-shaped plot bounded by a parabolic channel and straight boundaries. The boundaries are described by:

$$y = 4x - x^2, \quad x = 1, \quad x = 2, \quad \text{and the } x\text{-axis.}$$

Find the area enclosed by these curves.

(b) A water-tank designer asks you to derive the classical formula for the volume of a sphere by using solids of revolution. Consider the circle of radius r centered at the origin:

$$x^2 + y^2 = r^2.$$

Consider the portion of the circle in the first quadrant and, by rotating this quarter-area about the x -axis (or equivalently about the y -axis), set up the integral(s) and deduce the volume of a sphere of radius r .